Exploring uncertainty in Antarctica's future sea level contribution

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April 13, 2026

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Total word count: 2678 words.

Chapter 1

Literature Review

1.1 Sea level

1.1.1 Sea level and coastal communities

The IPCC's Sixth Assessment Report (AR6) highlights sea level rise as "an existential threat for island nations, low-lying coastal zones and the communities, infrastructure, and cultural heritage therein" (Cooley et al., 2022).

A rise of 1 m would increase the global population living below mean sea level from 34–49 million people today to 77–132 million, depending on the choice of digital elevation model and population dataset (Seeger and Minderhoud, 2026).

1.1.2 Global Mean Sea Level rise

1.2 The Antarctic Ice Sheet

The Antarctic ice sheet is the largest store of freshwater on Earth, holding **TODO: value** Gt of ice over an area of **TODO: value** km^2. It holds the potential to raise sea level by 57.25 m if it were to melt completely, although this is not considered likely in any future scenario of climate change (Fox-Kemper et al., 2021). Antarctica is situated over the South pole, bounded on all sides by the Southern Ocean. At its center, the ice sheet reaches altitudes of 4 km above sea level. This gives Antarctica one of the most extreme climates on Earth, with record minimum temperatures of -89.2°C (Turner et al., 2009). This section reviews the key regions of Antarctica, important changes to these regions in the recent past, and evidence of ice sheet change in the deep past (palaeoclimates).

1.2.1 Key regions of Antarctica

Antarctica is split into three climatically distinct regions: West Antarctica, East Antarctica, and the Antarctic Peninsula. The West Antarctic Ice Sheet (WAIS) is a 'marine' ice sheet, meaning that it is grounded on bedrock that lies predominantly below sea level. The deepest points of the bed are the Byrd subglacial basin and Bentley subglacial trench, reaching depths of 2.5 km below sea level. West Antarctic glaciers drain mainly into the Amundsen Sea, Ross Sea, and Weddell Sea sectors of the Southern Ocean. The latter two feature Antarctica's two largest floating ice shelves, the Ross ice shelf (**TODO: value** km^2) and the Filchner-Ronne ice shelf (**TODO: value** km^2), respectively. The ocean cavities beneath them are key regions for the production of dense deep waters that connect Antarctica to the rest of the world though the global ocean circulation

(Rintoul et al., 2025). **TODO: introduce Thwaites glacier here**.

The East Antarctic Ice Sheet (EAIS) currently stores the largest volume of ice in Antactica and represents the majority of Antarctica's potential to raise sea level. It is separated from West Antarctica by the Trans-Antarctic mountains, the highest of which is Mt Kirkpatrick at 4,528 m above sea level. It is largely a terrestrial ice sheet (grounded above sea level), but it has three key marine basins: The Recovery, Wilkes, and Aurora subglacial basins. Each has a sea level potential on the same order of magnitude as the entire West Antarctic Ice Sheet **TODO: cite**. East Antarctica's Amery ice shelf, the third largest ice shelf in Antarctica (**TODO: value** km^2), is the floating extension of the Lambert-Amery glacier system, Antarctica's largest drainage basin. East Antarctica is also home to Lake Vostok, the largest of the continent's subglacial lakes, with liquid freshwater that is up to 1 km deep (Siegert et al., 2001).

The Antarctic Peninsula Ice Sheet (APIS) is a comparatively small part of Antarctica, but is climatically distinct as the most Northerly region of the continent, reaching 68°S towards the Southern tip of South America. Its mountain range features smaller, Alpine-like glaciers down steep terrain, which contrast with the wide, low-gradient glaciers of the WAIS. The mountains act as a 2000 m orographic barrier between the Weddell Sea and the Bellinghausen Sea, blocking the prevailing Westerly winds and creating a steep East-West temperature and precipitation gradient across the peninsula (Schwerdtfeger, 1975; Van Lipzig et al., 2004). The largest ice shelf in the APIS, the Larsen ice shelf, extends down the Eastern flank of the peninsula and is divided into smaller sections Larsen A to D, the largest of which is Larsen C at **TODO: value and cite** km^2.

1.2.2 Recent observed changes

Antarctica is not in equilibrium with the current climate, with some regions showing a clear trend in mass balance that is consistent across multiple methods of measurement (Gardner et al., 2018; Rignot et al., 2019; Schröder et al., 2019; Shepherd et al., 2019; Smith et al., 2020; Velicogna et al., 2020; Wang et al., 2021) **NOTE: Admittedly, I plucked these from a 2022 review paper, but am struggling to find anything more recent. I could also perhaps trim them down to one of each: input/output, altimetry, gravimetry, or just cite IMBIE or IPCC**. The West Antarctic Ice Sheet is the main contributor to mass loss over this period, losing 82 ± 9 Gt yr^{-1} **NOTE: alternatively "contributing 6.6 mm to global mean sea level"** between 1992–2020 (Otosaka et al., 2023) **NOTE: acceleration?**. The vast majority of this mass loss has originated from the Amundsen Sea Sector of West Antarctica, particularly the Thwaites and Pine Island Glaciers, which in 2017 were losing mass at **TODO: values and cite Rignot 2019**, respectively. Sediment records from the Amundsen Sea indicate that Thwaites and Pine Island Glaciers have been in disequalibrium since at least the 1940s **TODO: cite**. Direct measurements of Pine Island Glacier made possible by the satellite era reveal a consistent pattern of change: retreat of the glacier terminus, acceleration of its ice stream, and a subsequent thinning as the glacier adjusts to maintain its flux balance (Joughin et al., 2003; Payne et al., 2004) **TODO: cite more recent**. It is difficult to draw direct links between the current disequilibrium and anthropogenic climate change, but it's likely that climate change has at least exacerbated the retreat of Pine Island glacier over this period (Bradley et al., 2025). **TODO: See Ines's 2023 remote sensing review paper for more inspiration on Pine Island**.

TODO: Thwaites

Unlike the Arctic, which has warmed by **TODO: value** times the global average due to nonlinear feedbacks (Previdi et al., 2021), Antarctic surface air temperatures have broadly followed the global trend **TODO: cite**. The one exception to this is the Antarctic Peninsula, which has warmed three times faster than the rest of the continent, up to temperatures otherwise unprecedented in the Holocene (Vaughan et al., 2003). This atmospheric warming has increased surface melting of the Antarctic Peninsula Ice Sheet and its resulting sea level contribution (Vaughan, 2006). The most pronounced changes in the APIS have been the thinning, retreat, and collapse of some of the peninsula's ice shelves (Cook and Vaughan, 2010; Rignot et al., 2013). The extent of ice shelves on the peninsula roughly trace the -9°C isotherm, suggesting a "limit of viability" for APIS ice shelves that will shrink with further warming (Morris and Vaughan, 2013). Although this is a useful heuristic, the mechanisms of ice shelf collapse are not purely thermal (see, for example, Sections 1.3.3 and 1.3.4).

The first ice shelf collapse to occur during the satellite era was the Wordie Ice Shelf in 1989 (Doake and Vaughan, 1991). This was followed by a sequential collapse from North to South down the Eastern flank of the peninsula, starting with the Larsen inlet — also in 1989 — then the Prince Gustav and Larsen A ice shelves in 1995 (Cook and Vaughan, 2010). Finally, the Larsen B ice shelf progressively thinned between 1992 and 2001 before its eventual collapse over a few days in February 2002 (Shepherd et al., 2003). The Larsen C ice shelf has so far avoided collapse, but has continued to thin at a rate of -3.8 m per decade over increasingly large areas of the ice shelf (Paolo et al., 2015). Between August 2015 and July 2017, a 200 km crack developed along the ice shelf, leading to the calving of a large iceberg representing 10% of the total ice shelf area (Hogg and Gudmundsson, 2017). Due to the loss of buttressing force provided by ice shelves (see Section 1.3.3), the collapse of each ice shelf on the peninsula was consistently followed by an acceleration and thinning of its upstream glaciers (Rignot et al., 2004; Glasser et al., 2011; Dømgaard et al., 2025). The collapse of the Prince Gustav and Larsen A ice shelves cannot be uniquely attributed to anthropogenic climate change, since sediment records indicate that they have previously collapsed during the mid-Holocene (Pudsey and Evans, 2001; Brachfeld et al., 2003). This is corroborated by written accounts of the Prince Gustav ice shelf by 19th century polar explorers suggesting that it had been in retreat since at least 1843 (Cooper, 1997). However, similar records from ocean sediments previously beneath the Larsen B ice shelf indicate that it has been stable throughout the Holocene, and therefore suggest that its recent collapse contains an anthropogenic signal (Domack et al., 2005).

NOTE: Too much on the Peninsula? Two big paragraphs on the smallest region (and least important for my work) seems like a lot. On the other hand, the peninsula is an excellent case study for what might happen elsewhere in Antarctica, an argument which I plan to develop in later sections, so it is perhaps worth going into detail here.

In contrast to the WAIS and APIS, sea level contribution from the East Antarctic Ice Sheet remains close to a state of balance, with a slight positive mass trend of 3 ± 15 Gt yr^{-1} between 1992–2020 (Otosaka et al., 2023). The high uncertainty in this estimate is the result of disagreement between different methods of measurement, with some studies showing a negative mass trend over similar periods (e.g. Rignot et al. 2019; Shepherd et al. 2019). These aggregate measurements also bely the spatial variability in East Antarctic mass trends. Several studies highlight regional mass loss from Wilkes Land in East Antarctica, particularly the Totten Glacier, which drains the Aurora subglacial basin (Shepherd et al., 2019; Velicogna et al., 2020; Smith et al., 2020). This is paired with a positive mass trend over the East Antarctic plataeu, consistent with reanalysis-derived increases in precipitation (Davis et al., 2005). This spatial variability suggest that whilst the East Antarctic Ice Sheet is in a state of balance, it is not in a state of equilibrium, with the potential for more significant mass change if this balance is disrupted.

TODO: Check Jamie's and Mira's theses to look for more EAIS notes.

TODO: Quick synthesis of this section?

1.2.3 Antarctica in past climates

NOTE: leaving this subsection for now because it's not clear to me how much detail to go into.

1.3 Ice sheet dynamics

1.3.1 Ice rheology*

NOTE: I may want a section here to talk about the polycrystalline structure of ice, as this is relevant to the flow law and the uncertain value of Glen's exponent, but perhaps this is getting too into the weeds.

1.3.2 Ice-atmosphere interactions*

1.3.3 Ice-ocean interactions

466 words

NOTE: I'm tying myself into knots trying to write this section. I can't make any sense of what the literature says, partly because oceanographers can't seem to make their minds up about how any of this works (or use any kind of consistent lexicon). What follows is a mess of half-written paragraphs.

The Southern Ocean is a confluence of water masses transported through the global ocean circulation. Deep water from the Pacific, Indian, and North Atlantic Oceans mix to form Circumpolar Deep Water (CDW), which circulates through the Antarctic Circumpolar Current (ACC) at depths of up to 3000 m (Talley, 2013). CDW is characterised generally by much warmer temperatures and higher salinity than the surface waters above it (Wang et al., 2025). It can be divided into ‘upper’ and ‘lower’ water masses with distinct characteristics. Upper CDW includes contributions from Indian and Pacific deep waters, which are warm, oxygen-poor, and nutrient-rich; Lower CDW contains more North Atlantic Deep Water, giving it characteristically high salinity (Whitworth et al., 2013). Upper CDW shoals as it moves Southwards, eventually mixing with surface waters at the Southern boundary of the ACC. However, lower CDW is dense enough to remain below the mixed layer and extend further towards the Antarctic continent (Orsi et al., 1995). The melting of Antarctic ice shelves is predominantly a question of whether this CDW is able to access the continental shelf that surrounds Antarctica (Wang et al., 2023).

Access to the shelf is limited by the Antarctic Slope Front (ASF), a density front along the shelf break, and a corresponding westward Antarctic Slope Current (ASC) (Jacobs, 1991). The ASF acts as a barrier between CDW and the cooler, fresher shelf waters that occupy ice shelf cavities around Antarctica. The structure of the ASF is highly variable across different sectors of Antarctica and is sensitive to local wind stresses (Spence et al., 2014) and freshwater fluxes (Si et al., 2023). These local conditions give rise to three distinct ice shelf cavity regimes: cold and fresh cavities, cold and saline cavities, and warm and saline cavities.

In Warm and saline cavities such as those in the Amundsen Sea, a weak ASF allows CDW to encroach onto the shelf through

The ASF is weakest in the Amundsen Sea, where the Eastward Antarctic Circumpolar Current dominates over the westward ASC (). This Eastward flow is associated with a polewards Ekman transport in the layer below, which helps to transport CDW onto the continental shelf (Klinck and Dinniman, 2010). This oc

The Southern boundary of the ACC flows very close to the continental shelf in the Amundsen and Bellinghausen Seas, but is deflected away from the Antarctic continent by cyclonic gyres in the Weddell and Ross Seas (Orsi et al., 1995). Temperatures in the gyres are generally lower than those found in ACC (Wang et al., 2023), but some heat is transported to the shelf break along the Eastern boundary of each gyre via entrained CDW (Assmann and Timmermann, 2005; Ryan et al., 2016).

have strengthened and shifted polewards as a response to greenhouse gas emissions and ozone depletion (Waugh et al., 2013; Lee and Feldstein, 2013).

1.3.4 Calving and hydrofracture*

1.3.5 Subglacial hydrology*

1.3.6 Ice sheet instability

1.3.7 Solid Earth effects

1.4 Ice sheet modelling

68 words

Numerical ice sheet models are important tools for Antarctic science, both for understanding the mechanisms behind recent ice sheet change (e.g. Payne et al. 2004; Gudmundsson et al. 2019; Bradley et al. 2025), exploring possible ice sheet configurations in the deep past (e.g. De Boer et al. 2015) and for making projections of future ice sheet change (Seroussi et al. 2020, 2024, among many others). This section reveiws the construction of a general ice sheet model and the important choices involved in simulating the Antarctic ice sheet.

1.4.1 Stokes flow

200 words

Practical ice flow modelling is done using a hierarchy of approximations to incompressible Stokes flow, which is itself the low Reynolds number limit of the Navier-Stokes equations. In this limit, inertial terms are ignored, and flow is governed by a balance between pressure gradients, viscous stresses, and gravitational driving stresses. These are solved alongside a constitutive relation, which defines the ice rheology (see Section 1.4.2). It is also necessary to impose boundary conditions at the base and surface of the ice sheet. For a rigid boundary (the interface at the bed), this might take the form:

$$\mathbf{u}_b = F(\boldsymbol{\tau}_b), \quad \mathbf{u}.\mathbf{n} = -m_b, \tag{1.1}$$

where $\mathbf{u}$ is the ice velocity field, $\mathbf{u}_b$ is the velocity component parallel to the bed, $\mathbf{n}$ is the normal to the bed, $\boldsymbol{\tau}_b$ is the basal shear stress, F is a sliding law (see Section 1.4.4), and m_b is the basal melt rate (in velocity units). For a free surface (the interface with the atmosphere or ocean), dynamic and kinematic boundary conditions are applied to specify the boundary stresses and surface evolution, respectively. Some model setups include an advection-diffusion equation for temperature, which couples to the momentum (Stokes) equations through the constitutive relation (Eq. 1.2). However, assuming isothermal ice (i.e. fixed temperature) is often a reasonable approximation over short timescales.

1.4.2 Constitutive relation

211 words

The constitutive relation describes the relationship between the stress and deformation of a material. For ice, it is common to use Glen's flow law (Glen, 1955):

$$\dot{\epsilon}_{ij} = A\tau^{n-1}\tau_{ij}, \quad \tau = \sqrt{\frac{1}{2}\tau_{ij}\tau_{ij}}, \tag{1.2}$$

where ϵ_{ij} is the strain rate tensor, τ_{ij} is the deviatoric stress tensor, $A(T)$ is a temperature-dependant rate factor, and n is a scalar exponent ("Glen's exponent"). Early work to estimate the value of n for polycrystalline ice suggested that $n \approx 3$ (Weertman 1983 and references therein). Subsequently, $n = 3$ became standard across a wide range of ice sheet models (e.g. Larour et al. 2012; Cornford et al. 2013; Lipscomb et al. 2019). However, Goldsby and Kohlstedt (2001) demonstrated that creep can be divided into multiple regimes, occuring through different mechanisms within the polycrystalline structure. The appropriate value of n depends on the dominant creep mechanism, which in turn depends on the stress. **NOTE: I could talk here about typical driving stresses in Antarctica and the corresponding value $n = 1.8$, but my attempts at this have gotten a bit bogged down and wordy, so have removed for better flow**. Other studies suggest that observations of Antarctic ice streams and ice shelves may be better explained by a higher value of n (e.g. Cuffey and Kavanaugh 2011; Millstein et al. 2022). Ice sheet models that have explored this found Antarctic sea level contribution to be sensitive to the choice of n, with higher values leading to greater sea level rise (Rosier et al., 2025; Getraer and Morlighem, 2025; Martin et al., 2026). **NOTE: Is signposting like "We will explore this uncertainty in chapter 2" (or similar) appropriate in these**

situations?

1.4.3 Approximations in ice flow models

39 words

Full Stokes modelling is computationally expensive and so is rarely deployed for continent-scale ice sheet modelling (with some notable exceptions, e.g. Seddik et al. 2012). Approximations are typically applied to match the requirements of a particular ice-dynamical regime.

1.4.4 Basal sliding

1.4.5 Surface mass balance*

1.4.6 Sub-shelf melting

1.4.7 Glacial isostatic adjustment

1.5 Projections of the Antarctic ice sheet

1.5.1 Century-scale projections

1.5.2 Millennium-scale projections*

1.5.3 Uncertainty Quantification

1.6 Uncertainty quantification*

1.6.1 Experiment design

1.6.2 Gaussian Process Emulation

1.6.3 Calibration

Chapter 2

Methods

This is the methods chapter.

Bibliography

Assmann, K. M. and Timmermann, R. (2005). Variability of dense water formation in the Ross Sea. *Ocean Dynamics*, 55(2):68–87.

Brachfeld, S., Domack, E., Kissel, C., Laj, C., Leventer, A., Ishman, S., Gilbert, R., Camerlenghi, A., and Eglinton, L. B. (2003). Holocene history of the Larsen-A Ice Shelf constrained by geomagnetic paleointensity dating. *Geology*, 31(9):749.

Bradley, A. T., Bett, D. T., Williams, C. R., Arthern, R. J., Holland, P. R., Bryne, J., and Edwards, T. L. (2025). Quantifying and attributing the role of anthropogenic climate change in industrial-era retreat of Pine Island Glacier.

Cook, A. J. and Vaughan, D. G. (2010). Overview of areal changes of the ice shelves on the Antarctic Peninsula over the past 50 years. *The Cryosphere*, 4(1):77–98.

Cooley, S., Schoeman, D., Bopp, L., Boyd, P., Donner, S., Ghebrehiwet, DY., Ito, S.-I., Kiessling, W., Martinetto, P., Ojea, E., Racault, M.-F., Rost, B., and Skern-Mauritzen, M. (2022). Oceans and Coastal Ecosystems and Their Services. In *Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, pages 379–550. H.-O. Pörtner, D.C. Roberts, M. Tignor, E.S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B. Rama (eds.)]. Cambridge University Press, Cambridge, UK and New York, NY, USA.

Cooper, A. (1997). Historical observations of Prince Gustav Ice Shelf. *Polar Record*, 33(187):285–294.

Cornford, S. L., Martin, D. F., Graves, D. T., Ranken, D. F., Le Brocq, A. M., Gladstone, R. M., Payne, A. J., Ng, E. G., and Lipscomb, W. H. (2013). Adaptive mesh, finite volume modeling of marine ice sheets. *Journal of Computational Physics*, 232(1):529–549.

Cuffey, K. and Kavanaugh, J. (2011). How nonlinear is the creep deformation of polar ice? A new field assessment. *Geology*, 39(11):1027–1030.

Davis, C. H., Li, Y., McConnell, J. R., Frey, M. M., and Hanna, E. (2005). Snowfall-Driven Growth in East Antarctic Ice Sheet Mitigates Recent Sea-Level Rise. *Science*, 308(5730):1898–1901.

De Boer, B., Dolan, A. M., Bernales, J., Gasson, E., Goelzer, H., Golledge, N. R., Sutter, J., Huybrechts, P., Lohmann, G., Rogozhina, I., Abe-Ouchi, A., Saito, F., and Van De Wal, R. S. W. (2015). Simulating the Antarctic ice sheet in the late-Pliocene warm period: PLISMIP-ANT, an ice-sheet model intercomparison project. *The Cryosphere*, 9(3):881–903.

Doake, C. S. M. and Vaughan, D. G. (1991). Rapid disintegration of the Wordie Ice Shelf in response to atmospheric warming. *Nature*, 350(6316):328–330.

Domack, E., Duran, D., Leventer, A., Ishman, S., Doane, S., McCallum, S., Amblas, D., Ring, J., Gilbert, R., and Prentice, M. (2005). Stability of the Larsen B ice shelf on the Antarctic Peninsula during the Holocene epoch. *Nature*, 436(7051):681–685.

Dømgaard, M., Millan, R., Andersen, J. K., Scheuchl, B., Rignot, E., Izeboud, M., Bernat, M., and Bjørk, A. A. (2025). Half a century of dynamic instability following the ocean-driven break-up of Wordie Ice Shelf. *Nature Communications*, 16(1):4016.

Fox-Kemper, B., Hewitt, HT., Xiao, C., Aðalgeirsdóttir, G., Drijfhout, SS., Edwards, TL., Golledge, NR., Hemer, M., Kopp, RE., Krinner, G., Mix, A., Notz, D., Nowicki, S., Nurhati, IS., Ruiz, L., Sallée, J.-B., Slangen, ABA., and Yu, Y. (2021). Ocean, Cryosphere and Sea Level Change. In *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, pages 1211–1362. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA.

Gardner, A. S., Moholdt, G., Scambos, T., Fahnstock, M., Ligtenberg, S., Van Den Broeke, M., and Nilsson, J. (2018). Increased West Antarctic and unchanged East Antarctic ice discharge over the last 7 years. *The Cryosphere*, 12(2):521–547.

Getraer, B. and Morlighem, M. (2025). Increasing the Glen–Nye Power-Law Exponent Accelerates Ice-Loss Projections for the Amundsen Sea Embayment, West Antarctica. *Geophysical Research Letters*, 52(7):e2024GL112516.

Glasser, N., Scambos, T., Bohlander, J., Truffer, M., Pettit, E., and Davies, B. (2011). From ice-shelf tributary to tidewater glacier: Continued rapid recession, acceleration and thinning of Röhss Glacier following the 1995 collapse of the Prince Gustav Ice Shelf, Antarctic Peninsula. *Journal of Glaciology*, 57(203):397–406.

Glen, J. W. (1955). The creep of polycrystalline ice. *Proceedings of the*

Royal Society of London. Series A. Mathematical and Physical Sciences, 228(1175):519–538.

Goldsby, D. L. and Kohlstedt, D. L. (2001). Superplastic deformation of ice: Experimental observations. *Journal of Geophysical Research: Solid Earth*, 106(B6):11017–11030.

Gudmundsson, G. H., Paolo, F. S., Adusumilli, S., and Fricker, H. A. (2019). Instantaneous Antarctic ice sheet mass loss driven by thinning ice shelves. *Geophysical Research Letters*, 46(23):13903–13909.

Hogg, A. E. and Gudmundsson, G. H. (2017). Impacts of the Larsen-C Ice Shelf calving event. *Nature Climate Change*, 7(8):540–542.

Jacobs, S. S. (1991). On the nature and significance of the Antarctic Slope Front. *Marine Chemistry*, 35(1-4):9–24.

Joughin, I., Rignot, E., Rosanova, C. E., Lucchitta, B. K., and Bohlander, J. (2003). Timing of Recent Accelerations of Pine Island Glacier, Antarctica: RECENT ACCELERATIONS OF PINE ISLAND GLACIER. *Geophysical Research Letters*, 30(13).

Klinck, J. M. and Dinniman, M. S. (2010). Exchange across the shelf break at high southern latitudes. *Ocean Science*, 6(2):513–524.

Larour, E., Seroussi, H., Morlighem, M., and Rignot, E. (2012). Continental scale, high order, high spatial resolution, ice sheet modeling using the Ice Sheet System Model (ISSM). *Journal of Geophysical Research: Earth Surface*, 117(F1):2011JF002140.

Lee, S. and Feldstein, S. B. (2013). Detecting Ozone- and Greenhouse Gas–Driven Wind Trends with Observational Data. *Science*, 339(6119):563–567.

Lipscomb, W. H., Price, S. F., Hoffman, M. J., Leguy, G. R., Bennett, A. R., Bradley, S. L., Evans, K. J., Fyke, J. G., Kennedy, J. H., Perego, M., Ranken, D. M., Sacks, W. J., Salinger, A. G., Vargo, L. J., and Worley, P. H. (2019). Description and evaluation of the Community Ice Sheet Model (CISM) v2.1. *Geoscientific Model Development*, 12(1):387–424.

Martin, D. F., Kachuck, S. B., Trevers, M., Millstein, J. D., Cornford, S. L., and Minchew, B. M. (2026). Rates of Sea-Level Rise Are Highly Sensitive to Ice Viscosity Parameters in Model Benchmarks. *AGU Advances*, 7(2):e2025AV001946.

Millstein, J. D., Minchew, B. M., and Pegler, S. S. (2022). Ice viscosity is more sensitive to stress than commonly assumed. *Communications Earth & Environment*, 3(1):57.

Morris, E. M. and Vaughan, D. G. (2013). Spatial and Temporal Variation of Surface Temperature on the Antarctic Peninsula And The Limit of Viability of Ice Shelves. In Domack, E., Levente, A., Burnet, A., Bindschadler, R., Convey, P., and Kirby, M., editors, *Antarctic Research Series*, pages 61–68. American Geophysical Union, Washington, D. C.

Orsi, A. H., Whitworth, T., and Nowlin, W. D. (1995). On the meridional extent and fronts of the Antarctic Circumpolar Current. *Deep Sea Research Part I: Oceanographic Research Papers*, 42(5):641–673.

Otosaka, I. N., Shepherd, A., Ivins, E. R., Schlegel, N.-J., Amory, C., Van Den Broeke, M. R., Horwath, M., Joughin, I., King, M. D., Krinner, G., Nowicki, S., Payne, A. J., Rignot, E., Scambos, T., Simon, K. M., Smith, B. E., Sørensen, L. S., Velicogna, I., Whitehouse, P. L., A, G., Agosta, C., Ahlstrøm, A. P., Blazquez, A., Colgan, W., Engdahl, M. E., Fettweis, X., Forsberg, R., Gallée, H., Gardner, A., Gilbert, L., Gourmelen, N., Groh, A., Gunter, B. C., Harig, C., Helm, V., Khan, S. A., Kittel, C., Konrad, H., Langen, P. L., Lecavalier, B. S., Liang, C.-C., Loomis, B. D., McMillan, M., Melini, D., Mernild, S. H., Mottram, R., Mouginot, J., Nilsson, J., Noël, B., Pattle, M. E., Peltier, W. R., Pie, N., Roca, M., Sasgen, I., Save, H. V., Seo, K.-W., Scheuchl, B., Schrama, E. J. O., Schröder, L., Simonsen, S. B., Slater, T., Spada, G., Sutterley, T. C., Vishwakarma, B. D., Van Wessem, J. M., Wiese, D., Van Der Wal, W., and Wouters, B. (2023). Mass balance of the Greenland and Antarctic ice sheets from 1992 to 2020. *Earth System Science Data*, 15(4):1597–1616.

Paolo, F. S., Fricker, H. A., and Padman, L. (2015). Volume loss from Antarctic ice shelves is accelerating. *Science*, 348(6232):327–331.

Payne, A. J., Vieli, A., Shepherd, A. P., Wingham, D. J., and Rignot, E. (2004). Recent dramatic thinning of largest West Antarctic ice stream triggered by oceans: THINNING OF LARGEST WEST ANTARCTIC ICE STREAM. *Geophysical Research Letters*, 31(23).

Previdi, M., Smith, K. L., and Polvani, L. M. (2021). Arctic amplification of climate change: A review of underlying mechanisms. *Environmental Research Letters*, 16(9):093003.

Pudsey, C. J. and Evans, J. (2001). First survey of Antarctic sub–ice shelf sediments reveals mid-Holocene ice shelf retreat. *Geology*, 29(9):787.

Rignot, E., Casassa, G., Gogineni, P., Krabill, W., Rivera, A., and Thomas, R. (2004). Accelerated ice discharge from the Antarctic Peninsula following the collapse of Larsen B ice shelf. *Geophysical Research Letters*, 31(18):2004GL020697.

Rignot, E., Jacobs, S., Mouginot, J., and Scheuchl, B. (2013). Ice-Shelf Melting Around Antarctica. *Science*, 341(6143):266–270.

Rignot, E., Mouginot, J., Scheuchl, B., Van Den Broeke, M., Van Wessem, M. J., and Morlighem, M. (2019). Four decades of Antarctic Ice Sheet mass balance from 1979–2017. *Proceedings of the National Academy of Sciences*, 116(4):1095–1103.

Rintoul, S. R., Stewart, A. L., Johnson, G. C., Zhou, S., Foppert, A., Li, Q., Morrison, A. K., Silvano, A., Gunn, K. L., England, M. H., Nihashi, S., and Aoki, S. (2025). Antarctic Bottom Water in a changing climate. *Nature Reviews Earth & Environment*, 7(2):86–102.

Rosier, S. H. R., Gudmundsson, G. H., Jenkins, A., and Naughten, K. A. (2025). Calibrated sea level contribution from the Amundsen Sea sector, West Antarctica, under RCP8.5 and Paris 2C scenarios. *The Cryosphere*, 19(7):2527–2557.

Ryan, S., Schröder, M., Huhn, O., and Timmermann, R. (2016). On the warm inflow at the eastern boundary of the Weddell Gyre. *Deep Sea Research Part I: Oceanographic Research Papers*, 107:70–81.

Schröder, L., Horwath, M., Dietrich, R., Helm, V., Van Den Broeke, M. R., and Ligtenberg, S. R. M. (2019). Four decades of Antarctic surface elevation changes from multi-mission satellite altimetry. *The Cryosphere*, 13(2):427–449.

Schwerdtfeger, W. (1975). The Effect of the Antarctic Peninsula on the Temperature Regime of the Weddell Sea. *Monthly Weather Review*, 103(1):45–51.

Seddik, H., Greve, R., Zwinger, T., Gillet-Chaulet, F., and Gagliardini, O. (2012). Simulations of the Greenland ice sheet 100 years into the future with the full Stokes model Elmer/Ice. *Journal of Glaciology*, 58(209):427–440.

Seeger, K. and Minderhoud, P. S. J. (2026). Sea level much higher than assumed in most coastal hazard assessments. *Nature*.

Seroussi, H., Nowicki, S., Payne, A. J., Goelzer, H., Lipscomb, W. H., Abe-Ouchi, A., Agosta, C., Albrecht, T., Asay-Davis, X., Barthel, A., Calov, R., Cullather, R., Dumas, C., Galton-Fenzi, B. K., Gladstone, R., Golledge, N. R., Gregory, J. M., Greve, R., Hattermann, T., Hoffman, M. J., Humbert, A., Huybrechts, P., Jourdain, N. C., Kleiner, T., Larour, E., Leguy, G. R., Lowry, D. P., Little, C. M., Morlighem, M., Pattyn, F., Pelle, T., Price, S. F., Quiquet, A., Reese, R., Schlegel, N.-J., Shepherd, A., Simon, E., Smith, R. S., Straneo, F., Sun, S., Trusel, L. D., Van Breedam, J., van de Wal, R. S. W., Winkelmann, R., Zhao, C., Zhang, T., and Zwinger, T. (2020). ISMIP6 Antarctica: A multi-model ensemble of the Antarctic ice sheet evolution over the 21st century. *The Cryosphere*, 14(9):3033–3070.

Seroussi, H., Pelle, T., Lipscomb, W. H., Abe-Ouchi, A., Albrecht, T., Alvarez-Solas, J., Asay-Davis, X., Barre, J.-B., Berends, C. J., Bernales, J., Blasco, J., Caillet, J., Chandler, D. M., Coulon, V., Cullather, R., Dumas, C., Galton-Fenzi, B. K., Garbe, J., Gillet-Chaulet, F., Gladstone, R., Goelzer, H., Golledge, N., Greve, R., Gudmundsson, G. H., Han, H. K., Hillebrand, T. R., Hoffman, M. J., Huybrechts, P., Jourdain, N. C., Klose, A. K., Langebroek, P. M., Leguy, G. R., Lowry, D. P., Mathiot, P., Montoya, M., Morlighem, M., Nowicki, S., Pattyn, F., Payne, A. J., Quiquet, A., Reese, R., Robinson, A., Saraste, L., Simon, E. G., Sun, S., Twarog, J. P., Trusel, L. D., Urruty, B., Van Breedam, J., Van De Wal, R. S. W., Wang, Y., Zhao, C., and Zwinger, T. (2024). Evolution of the Antarctic Ice Sheet Over the Next Three Centuries From an ISMIP6 Model Ensemble. *Earth's Future*, 12(9):e2024EF004561.

Shepherd, A., Gilbert, L., Muir, A. S., Konrad, H., McMillan, M., Slater, T., Briggs, K. H., Sundal, A. V., Hogg, A. E., and Engdahl, M. E. (2019). Trends in Antarctic Ice Sheet Elevation and Mass. *Geophysical Research Letters*, 46(14):8174–8183.

Shepherd, A., Wingham, D., Payne, T., and Skvarca, P. (2003). Larsen Ice Shelf Has Progressively Thinned. *Science*, 302(5646):856–859.

Si, Y., Stewart, A. L., and Eisenman, I. (2023). Heat transport across the Antarctic Slope Front controlled by cross-slope salinity gradients. *Science Advances*, 9(18):eadd7049.

Siegert, M. J., Ellis-Evans, J. C., Tranter, M., Mayer, C., Petit, J.-R., Salamatin, A., and Priscu, J. C. (2001). Physical, chemical and biological processes in Lake Vostok and other Antarctic subglacial lakes. *Nature*, 414(6864):603–609.

Smith, B., Fricker, H. A., Gardner, A. S., Medley, B., Nilsson, J., Paolo, F. S., Holschuh, N., Adusumilli, S., Brunt, K., Csatho, B., Harbeck, K., Markus, T., Neumann, T., Siegfried, M. R., and Zwally, H. J. (2020). Pervasive ice sheet mass loss reflects competing ocean and atmosphere processes. *Science*, 368(6496):1239–1242.

Spence, P., Griffies, S. M., England, M. H., Hogg, A. M., Saenko, O. A., and Jourdain, N. C. (2014). Rapid subsurface warming and circulation changes of Antarctic coastal waters by poleward shifting winds. *Geophysical Research Letters*, 41(13):4601–4610.

Talley, L. D. (2013). Closure of the Global Overturning Circulation Through the Indian, Pacific, and Southern Oceans: Schematics and Transports. *Oceanography*, 26:80–97.

Turner, J., Anderson, P., Lachlan-Cope, T., Colwell, S., Phillips, T., Kirchgaessner, A., Marshall, G. J., King, J. C., Bracegirdle, T., Vaughan, D. G., Lagun, V., and Orr, A. (2009). Record low surface air temperature at Vostok station, Antarctica. *Journal of Geophysical Research: Atmospheres*, 114(D24):2009JD012104.

Van Lipzig, N. P. M., King, J. C., Lachlan-Cope, T. A., and Van Den Broeke, M. R. (2004). Precipitation, sublimation, and snow drift in the Antarctic Peninsula region from a regional atmospheric model. *Journal of Geophysical Research: Atmospheres*, 109(D24):2004JD004701.

Vaughan, D. G. (2006). Recent Trends in Melting Conditions on the Antarctic Peninsula and Their Implications for Ice-sheet Mass Balance and Sea Level. *Arctic, Antarctic, and Alpine Research*, 38(1):147–152.

Vaughan, D. G., Marshall, G. J., Connolley, W. M., Parkinson, C., Mulvaney, R., Hodgson, D. A., King, J. C., Pudsey, C. J., and Turner, J. (2003). Recent Rapid Regional Climate Warming on the Antarctic Peninsula. *Climatic Change*, 60(3):243–274.

Velicogna, I., Mohajerani, Y., A, G., Landerer, F., Mouginot, J., Noel, B., Rignot, E., Sutterley, T., Van Den Broeke, M., Van Wessem, M., and Wiese, D. (2020). Continuity of Ice Sheet Mass Loss in Greenland and Antarctica From the GRACE and GRACE Follow-On Missions. *Geophysical Research Letters*, 47(8).

Wang, L., Davis, J. L., and Howat, I. M. (2021). Complex Patterns of Antarctic Ice Sheet Mass Change Resolved by Time-Dependent Rate Modeling of GRACE and GRACE Follow-On Observations. *Geophysical Research Letters*, 48(1):e2020GL090961.

Wang, Z., Li, Y., Shi, J., Zhang, Z., Liu, C., Zhou, M., and Wei, Z. (2025). Water masses in the Southern Ocean: Variability, trends, and drivers. *Acta Oceanologica Sinica*, 44(3):35–56.

Wang, Z., Liu, C., Cheng, C., Qin, Q., Yan, L., Qian, J., Sun, C., and Zhang, L. (2023). On the Multiscale Oceanic Heat Transports Toward the Bases of the Antarctic Ice Shelves. *Ocean-Land-Atmosphere Research*, 2:0010.

Waugh, D. W., Primeau, F., DeVries, T., and Holzer, M. (2013). Recent Changes in the Ventilation of the Southern Oceans. *Science*, 339(6119):568–570.

Weertman, J. (1983). CREEP DEFORMATION OF ICE. *Annual Review of Earth and Planetary Sciences*, 11(1):215–240.

Whitworth, T., Orsi, A. H., Kim, S.-J., Nowlin, W. D., and Locarnini, R. A. (2013). Water Masses and Mixing Near the Antarctic Slope Front. In Jacobs, S. S. and Weiss, R. F., editors, *Antarctic Research Series*, pages 1–27. American Geophysical Union, Washington, D. C.